

Question 3 (Composite Functions & Inverse Trigonometry)

Problem Statement

If $u = \sin^{-1} \left[\frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} \right]$, prove that:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{20} \tan u$$

Step-by-Step Solution

Step 1: Transform the Function to Remove the Outer Non-Homogeneous Wrapper

Because of the outer inverse sine function, u itself is not homogeneous. To eliminate this wrapper, take the sine of both sides to create a new, purely algebraic substitute function, z :

$$z = \sin u = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}}$$

Step 2: Determine the Degree (n) of the Substitute Function z

Define the function explicitly:

$$f(x, y) = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}}$$

Substitute $x \rightarrow xt$ and $y \rightarrow yt$ into the equation:

$$f(xt, yt) = \frac{(xt)^{1/4} + (yt)^{1/4}}{(xt)^{1/5} + (yt)^{1/5}}$$

Distribute the exponential fractional powers:

$$f(xt, yt) = \frac{x^{1/4} \cdot t^{1/4} + y^{1/4} \cdot t^{1/4}}{x^{1/5} \cdot t^{1/5} + y^{1/5} \cdot t^{1/5}}$$

Factor out $t^{1/4}$ from the numerator and $t^{1/5}$ from the denominator:

$$f(xt, yt) = \frac{t^{1/4} (x^{1/4} + y^{1/4})}{t^{1/5} (x^{1/5} + y^{1/5})}$$

Simplify the powers of t using the subtraction rule for exponents ($\frac{a^m}{a^n} = a^{m-n}$):

$$n = \frac{1}{4} - \frac{1}{5} = \frac{5-4}{20} = \frac{1}{20}$$

$$f(xt, yt) = t^{1/20} \left(\frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}} \right) = t^{1/20} z$$

This confirms that the substitute function z is a homogeneous function with a degree of $n = \frac{1}{20}$.

Step 3: Apply Euler's Theorem to z

By Euler's Theorem, applying the differential operators to our homogeneous function z gives:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz$$

Substitute the calculated degree $n = \frac{1}{20}$:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{20}z \quad \text{--- (Equation 1)}$$

Step 4: Convert Partial Derivatives of z Back to u via Chain Rule

Since $z = \sin u$, and u depends on both x and y , calculate the partial derivatives of z using the chain rule:

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \cos u \cdot \frac{\partial u}{\partial x} \\ \frac{\partial z}{\partial y} &= \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = \cos u \cdot \frac{\partial u}{\partial y} \end{aligned}$$

Substitute these chain-rule expressions directly back into the left-hand side of Equation 1:

$$x \left(\cos u \cdot \frac{\partial u}{\partial x} \right) + y \left(\cos u \cdot \frac{\partial u}{\partial y} \right) = \frac{1}{20}z$$

Factor out the common term $\cos u$:

$$\cos u \cdot \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = \frac{1}{20}z$$

Substitute our original definition of z ($z = \sin u$) into the right-hand side:

$$\cos u \cdot \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = \frac{1}{20} \sin u$$

Divide both sides by $\cos u$ to isolate our target expression:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{20} \left(\frac{\sin u}{\cos u} \right)$$

Using the trigonometric identity $\frac{\sin u}{\cos u} = \tan u$:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{20} \tan u = \text{RHS}$$

Hence, proved.